Applied Data Science (52774)

Module-Wise Question Bank (Previous Year Papers)

Notes: Freq = number of times the question appeared across 5 papers. Marks = marks assigned (x/y = different papers assigned different marks).

Module 1 – Introduction to Data Science

Topics covered: *Introduction to Data Science, Data Science Process, Motivation (Volume/Dimensions/Complexity), Data Science Tasks, Data Preparation & Modeling, Difference between Data Science and Data Analytics*

Question	Freq	Marks
What is the difference between data science and data analytics?	3	5/10
Explain the data science process in detail.	3	10
Explain the data science tasks with proper examples.	1	10
Explain the significance of data science considering Volume and Dimensions of Data.	1	10
Explain the significance of Volume, Dimension and Complexity to Data Science Techniques.	1	10

Module 2 – Data Exploration

Topics covered: *Types/Properties of Data, Descriptive Statistics, Univariate & Multivariate Exploration, Measures of Central Tendency/Spread, Symmetry, Skewness (Karl Pearson, Bowley), Kurtosis, Correlation (Karl Pearson), Inferential Statistics, Normal/Poisson Distributions, Hypothesis Testing, Central Limit Theorem, Confidence Interval, Z-test, t-test, Type I/II Errors, ANOVA*

Question	Freq	Marks
What are Type I and Type II errors? Give examples.	3	5/10
Write a note on Univariate Exploration and Multivariate Exploration.	2	10
Calculate the coefficient of correlation using Karl Pearson's method. [X: 10,20,30,40,50,60,70,80,90,100 | Y: 2,4,8,5,10,15,14,20,22,50]	2	10
Calculate the coefficient of correlation using Karl Pearson's method. [X: 15,18,20,28,34 | Y: 40,42,46,50,52]	1	10
Find the correlation between advertisement expenses and sales volume for 10 companies using Karl Pearson's coefficient. [Company: 1-10 | Advt: 11,13,14,16,16,15,15,14,13,13 | Sales: 50,50,55,60,65,65,65,60,60,50]	1	10
Find Bowley's coefficient of skewness. [Size: 4,4.5,5,5.5,6,6.5,7,7.5,8 | F: 10,18,22,25,40,15,10,8,7]	2	10
Find Bowley's coefficient of skewness. [Profit(crores): 4-8,8-12,12-16,16-20,20-24 | No. of films: 4,10,15,8,3]	2	10
Find Bowley's coefficient of skewness for the series:	1	10

2,4,6,8,10,12,14,16,18,20,22		
Find coefficient of skewness. [Size: 3,4,5,6,7,8,9,10 | Frequency: 7,10,14,35,102,136,43,8]	1	10
Calculate Bowley's coefficient of skewness. [Income: 30-40,40-50,50-60,60-70,70-80,80-90,90-100 | No.of persons: 8,24,48,68,30,13,9]	1	10
Find the coefficient of skewness using Bowley's method. [Salary: 4-8,8-12,12-16,16-20,20-24 | No. of Candidates: 4,10,15,8,3]	1	10
Brief about ANOVA and its types. How is it different from a t-test?	2	5/10
What is ANOVA? Brief about benefits of ANOVA technique.	1	10
What is Hypothesis testing? Explain the steps involved in Hypothesis testing with an example.	2	10
What is Hypothesis testing? Write about the different types of Hypothesis testing.	1	10
Examine the significance of weight increase in children due to food B using t-test. [Food A: 49,53,51,52,47,50,52,53 | Food B: 52,55,52,53,50,54,54,53] (t-value at alpha=0.05 is 2.365)	1	10
A stenographer claims she can type 120 wpm. Can we reject her claim based on 100 trials with mean=116 and SD=15? (Z_a=1.96, 5% significance)	1	10
Write a note on measure of spread.	1	5

Module 3 – Methodology and Data Visualization

Topics covered: *Model Building Overview, Cross Validation, K-fold Cross Validation, Leave-1-out, Bootstrapping, Univariate & Multivariate Visualization (Histogram, Quartile, Distribution Chart, Scatter Plot, Scatter Matrix, Bubble Chart, Density Chart), Roadmap for Data Exploration, High-dimensional Visualization (Parallel, Deviation, Andrews Curves)*

Question	Freq	Marks
Describe the terms: Cross-validation, K-fold cross-validation, leave-1-out and Bootstrapping.	4	10
Explain Cross Validation and K-fold cross validation in detail.	1	10
Explain validation techniques: Bootstrap and Cross-validation.	1	5
What is Data Visualization? Explain Multivariate data with a Scatter plot example.	1	10
Write a note on Data Visualization techniques.	2	10
State the importance of Data Visualization. State the purpose of scatter plots, quartile plots, bubble charts, density chart.	2	5/10
Explain in brief the objectives of Data Exploration.	2	5

Module 4 – Anomaly Detection

Topics covered: *Outliers, Causes of Outliers, Anomaly Detection Techniques, Outlier Detection using Statistics, Distance-based Methods, Density-based Methods, SMOTE*

Question	Freq	Marks
What are outliers? Explain different outlier detection methods.	3	10
Explain the density-based outlier detection approach (DBSCAN).	2	10
Explain the DBSCAN algorithm to detect outliers. Give the advantages and disadvantages.	1	10
What are outliers and their causes? Can statistics be used to detect outliers? Explain.	1	10
Brief about SMOTE.	3	5/10
Explain SMOTE in detail.	3	10
What is anomaly detection? Explain the process of anomaly detection.	1	5
State the reasons for outliers occurring in a dataset.	2	5

Module 5 – Time Series Forecasting

Topics covered: *Taxonomy of Time Series Forecasting Methods, Time Series Decomposition, Smoothening Methods (Average, Moving Average), Time Series Analysis using Linear Regression, ARIMA Model, Performance Evaluation (MAE, RMSE, MAPE, MASE)*

Question	Freq	Marks
What do you mean by Time Series Decomposition?	1	5
Explain in brief the taxonomy of time series forecasting methods.	3	5/10
What are the attributes of time series decomposition? Explain the classical decomposition technique.	1	10
Explain the Auto Regressive Integrated Moving Average (ARIMA) model in detail.	2	10
What are the pros and cons of an ARIMA model? Explain with proper examples.	1	10
Explain the working of the ARIMA model in detail.	1	10
Explain smoothing methods in Time Series Forecasting.	1	10
Describe how the time-series approach is useful for forecasting the demand for a product.	2	10
Explain how the time-series approach is used to forecast the demand for a product.	1	10
Explain performance evaluation with respect to Time Series Forecasting.	1	5

Module 6 – Applications of Data Science

Topics covered: *Predictive Modeling (House Price Prediction, Fraud Detection), Clustering (Customer Segmentation), Time Series Forecasting (Weather Forecasting), Recommendation Engines (Product Recommendation)*

Question	Freq	Marks
Explain predictive modelling applied to House Price Prediction.	3	10
Write a note on House Price Prediction or Fraud Detection.	1	10
Explain how to build a product recommendation model in detail.	1	10
What are Recommendation engines? Explain.	1	10
Write a note on Applications of Data Science.	1	10
Time series analysis using linear regression.	1	10

Unclassified / Multi-Module Questions

Topics covered: *Questions spanning multiple modules or not clearly mapped to a single module*

Question	Freq	Marks
Write a note on any TWO: (i) Data Visualization techniques (ii) Univariate & Multivariate Exploration (iii) House Price Prediction or Fraud Detection	1	20
Write a note on any FOUR: (A) Data Visualization (B) Applications of Data Science (C) Data Exploration (D) Taxonomy of time series forecasting (E) Outlier detection methods	1	20
Write a note on any TWO: (i) Data Visualization (ii) Univariate & Multivariate Exploration (iii) Fraud Detection	1	20